Q1

Considering a computer system with a 10-bit logical address. Translate the logical address 1000011011 to the corresponding physical address for the following two cases:

a) Assuming paging is used, the page size is 128 bytes and the page table is as given in Figure Q4a.

b) Assuming segmentation is used, the maximum segment size that the system can support is 256 bytes, and the segment table is as given in Figure Q4b.

0	01011
1	00010
2	00010
3	10101
4	01001
5	01100
6	11010
7	00110

Figure Q4a

0	00010000	010000000000
1	00001000	100000000000
2	00100000	010011010000
3	00011000	100001100000

Figure Q4b

Q1 (a)

- Given page size is 128 bytes→ **7** bits are needed for page offset and **3** bits for page number.

$$\underbrace{100}_{\text{Page number}}\underbrace{0011011}_{\text{offset}}$$

0	01011
1	00010
2	00010
3	10101
4	01001
5	01100
6	11010
7	00110

Page table

$$\Rightarrow \underline{01001}\mathbf{0011011}$$

Q1 (b)

- The maximum segment size is 256 bytes→ **8** bits are needed for the segment offset and **2** bits for segment number.

$\underbrace{10}_{\text{Seg. number}}\underbrace{00011011}_{\text{offset}}$

0	00010000	010000000000
1	00001000	100000000000
2	00100000	010011010000
3	00011000	100001100000

Segment table

00011011 (offset)

\+ 010011010000 (base)

= 010011101011

Q2

Assume a computer system uses paging for memory allocation. Logical address consists of *x* bits for the page number and *y* bits for the offset.

a) What is the page size?

a) How many entries are there in the page table?

a) Assuming each page table entry occupies 4 bytes, how many entries are there in the outer level page table if a two-level paging scheme is used? What is the format of logical address?

Q2 (Answer)

a) Page size: 2^y bytes

b) Number of entries in page table: 2^x entries

c) Number of entries in the outer level page table: $(2^x \times 4)/2^y = 2^{x-y} \times 4$

Number of entries per page in a page table: $2^y/4$

Logical address format: x-y+2 | y-2 | y

Q3

For each of the page replacement policies listed below, calculate the number of page faults encountered when referencing the following pages. Assume the availability of 4 empty page frames.

0 1 6 0 3 4 0 1 0 3 4 6 3 4

a) FIFO

b) CLOCK

c) LRU

Q3 (FIFO)

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	4	4	4	4	4	4	4	3	3
F2		1	1	1	1	1	0	0	0	0	0	0	0	4
F3			6	6	6	6	6	1	1	1	1	1	1	1
F4					3	3	3	3	3	3	3	6	6	6
						0	1	6				3	4	0
	P	P	P		P	P	P	P				P	P	P

There are 10 page faults.

Q3 (CLOCK)

Page Number

R Bit

Circular Queue

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0,0	0,0	0,0	0,1	0,1	0,0	0,1	0,1	0,1	0,1	0,1	0,0	0,0	0,0
F2		1,0	1,0	1,0	1,0	4,0	4,0	4,0	4,0	4,0	4,1	4,0	4,0	4,1
F3			6,0	6,0	6,0	6,0	6,0	1,0	1,0	1,0	1,0	6,0	6,0	6,0
F4					3,0	3,0	3,0	3,0	3,0	3,1	3,1	3,0	3,1	3,1
						1		6				1		
	P	P	P		P	P		P				P		

There are 7 pages faults.

Clock hand in shaded entry.

Q3 (LRU)

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F2		1	1	1	1	4	4	4	4	4	4	4	4	4
F3			6	6	6	6	6	1	1	1	1	6	6	6
F4					3	3	3	3	3	3	3	3	3	3
						1		6				1		
	P	P	P		P	P		P				P		

There are 7 page faults.